

In Exercises 17-20, show that T is a linear transformation by finding matrix that implements the mapping. Note that $x_1, x_2, \dots$ are not vectors but are entries in vectors.

17. $T(x_1, x_2, x_3, x_4) = (0, x_1 + x_2, x_2 + x_3, x_3 + x_4)$

Solve

$$\begin{bmatrix} 0 \\ x_1 + x_2 \\ x_2 + x_3 \\ x_3 + x_4 \end{bmatrix} = \begin{bmatrix} 0 \cdot x_1 \\ 1 \cdot x_1 \\ 0 \cdot x_1 \\ 0 \cdot x_1 \end{bmatrix} + \begin{bmatrix} 0 \cdot x_2 \\ 1 \cdot x_2 \\ 1 \cdot x_2 \\ 0 \cdot x_2 \end{bmatrix} + \begin{bmatrix} 0 \cdot x_3 \\ 0 \cdot x_3 \\ 1 \cdot x_3 \\ 1 \cdot x_3 \end{bmatrix} + \begin{bmatrix} 0 \cdot x_4 \\ 0 \cdot x_4 \\ 0 \cdot x_4 \\ 1 \cdot x_4 \end{bmatrix} \quad (\text{vector addition})$$

$T(x)$

$$= x_1 \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} + x_4 \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \quad (\text{scalar multiplication})$$

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$$= \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \quad (\text{By defn. of matrix eqn. } Ax=b)$$

By Theorem 10, the standard matrix of T is $\begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} \quad \square$

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19. $T(x_1, x_2, x_3) = (x_1 - 5x_2 + 4x_3, x_2 - 6x_3)$

$$\begin{bmatrix} x_1 - 5x_2 + 4x_3 \\ x_2 - 6x_3 \end{bmatrix} = \begin{bmatrix} x_1 \\ 0 \cdot x_1 \end{bmatrix} + \begin{bmatrix} -5x_2 \\ 0 \cdot x_2 \end{bmatrix} + \begin{bmatrix} 4x_3 \\ -6x_3 \end{bmatrix} \quad (\text{vector addition})$$

$$= x_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} -5 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 4 \\ -6 \end{bmatrix} \quad (\text{scalar multiplication})$$

$$= \begin{bmatrix} 1 & -5 & 4 \\ 0 & 0 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad (\text{By defn. of matrix eqn. } Ax=b)$$

By Theorem 10, the standard matrix of T is $\begin{bmatrix} 1 & -5 & 4 \\ 0 & 0 & -6 \end{bmatrix} \quad \square$

21. Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation such that $T(x_1, x_2) = (x_1 + x_2, 4x_1 + 5x_2)$. Find x such that $T(x) = (3, 8)$

Solution: $\begin{bmatrix} x_1 + x_2 \\ 4x_1 + 5x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 8 \end{bmatrix} \Rightarrow \begin{bmatrix} x_1 \\ 4x_1 \end{bmatrix} + \begin{bmatrix} x_2 \\ 5x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 8 \end{bmatrix} \quad (\text{vector addition})$

$$\Rightarrow x_1 \begin{bmatrix} 1 \\ 4 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 5 \end{bmatrix} = \begin{bmatrix} 3 \\ 8 \end{bmatrix} \quad (\text{scalar multiplication})$$